

LPIN Communication Scale Test Bed

Benchmark Scorecard · Full Exercise P0–P6

Generated 2026-07-31 17:06 UTC · 56 projects · 35 US states (70%) · Local fidelity only · Not a city portal

PROJECTS 56/56 all gates pass	MESSAGES 2,688 suite field comms	NATIONALS 28 multi-project vendors
MSGS P50 48 floor 40	ROUTING GAPS 0 critical path clean	STATES 35 70% of US

Verdict

All seven phase gates (P0–P6) are **OPEN** at **56/56** projects. The suite is a viable device-local baseline for streamlined permitting, scheduling, and operations communication simulation. Private chat lab remains pattern source only — not wholesale production merge.

Phase gates

Phase	Marker	Pass	Fail	Gate	Duration
P0	[P0-BASE]	56	0	OPEN	1 ms
P1	[P1-ORG]	56	0	OPEN	14 ms
P2	[P2-COMM]	56	0	OPEN	27 ms
P3	[P3-VENDOR]	56	0	OPEN	33 ms
P4	[P4-NATL]	56	0	OPEN	12 ms
P5	[P5-DEPTH]	56	0	OPEN	86 ms
P6	[P6-SCALE]	56	0	OPEN	69 ms

Suite inventory

Industry	Projects
multi_family	8
civil	7
industrial	7
commercial	6
education	6

Industry	Projects
hospitality	6
healthcare	5
renovation	5
single_family	5
Seed + demo total	56

Scale ops (P6) benchmarks

Metric	Value
Permit coverage	100%
Schedule coverage	100%
Total field messages	2,688
Msgs p50 / p95	48 / 48
Routing gaps	0
Open acks (tracked)	143
National vendors	28
Avg projects / national	15.96
Project national coverage	100%

Phase suite checks (detail)

P0 [P0-BASE] · **OPEN** · 56/56 pass

ID	OK	Detail
P0.S1	●	count=56 target=56
P0.S2	●	states=35 (70% US)
P0.S3	●	ids unique

P1 [P1-ORG] · **OPEN** · 56/56 pass

ID	OK	Detail
P1.S1	●	org pass 56/56
P1.S2	●	avg divisions=13.1
P1.S3	●	avg contracts=7.4

P2 [P2-COMM] · **OPEN** · 56/56 pass

ID	OK	Detail
P2.S1	●	comm pass 56/56
P2.S2	●	msgs p50=48 p95=48 total=2688
P2.S3	●	routing_gaps=0

P3 [P3-VENDOR] · **OPEN** · 56/56 pass

ID	OK	Detail
P3.S1	●	vendor-comm pass 56/56
P3.S2	●	avg vendors in log=5.5
P3.S3	●	avg contracts in log=6.5

P4 [P4-NATL] · **OPEN** · 56/56 pass

ID	OK	Detail
P4.1	●	national vendors=28
P4.2	●	all nationals span ≥ 2 projects
P4.3	●	all vendors have reg/quality/procurement affinity
P4.4	●	ledger vendors=28 projects_indexed=56
P4.5	●	project coverage 100% (56/56)
P4.S1	●	registry size=28
P4.S2	●	avg projects/vendor=15.96
P4.S3	●	coverage=100%

P5 [P5-DEPTH] · **OPEN** · 56/56 pass

ID	OK	Detail
P5.S1	●	depth pass 56/56
P5.S2	●	msgs min=48 p50=48 p95=48 total=2688

P6 [P6-SCALE] · **OPEN** · 56/56 pass

ID	OK	Detail
P6.1	●	permit coverage 100% (56/56)
P6.2	●	schedule coverage 100% (56/56)
P6.3	●	routing_gaps=0 openAcks=143
P6.4	●	comms total=2688 p50=48 p95=48
P6.5	●	archive written by harness (see docs/test-runs)
P6.S1	●	project pass 56/56

Industry pass mix (P6 boards)

Industry	N	Pass	Msgs (sum)
civil	7	7	336
commercial	6	6	288
education	6	6	288
healthcare	5	5	240
hospitality	6	6	288
industrial	7	7	336
multi_family	9	9	432
renovation	5	5	240
single_family	5	5	240

Fidelity notes

- Synthetic people, agencies, vendors only — no real municipality PII.
- Device-local packs; geometry never overrides AHJ state packs.
- Private chat lab is pattern source; not production deploy.
- Multi-window review uses locked panes (no sync thrash).
- Reproduce: `npm run test:portfolio:gates`

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